Regime-Dependent Predictive Structure Between Equity Factors: Evidence from Granger Causality

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Abstract

We evaluate regime-dependent predictive structure between equity factors using 35 years of Fama-French daily data (1990–2024) and a Student-t Hidden Markov Model with 50 multi-start initializations. The primary regime fit is selected by leakage-safe likelihood-only ranking; a 2008-screened fit is reported as sensitivity only. In the primary fit, HML→SMB is strongly significant in the Normal regime at lag 1 ($p = 8.75 \times 10^{-9}$; HAC $p = 9.45 \times 10^{-8}$). Elevated-regime evidence is directional but below the paper's confirmatory multiple-testing threshold ($p = 0.004$ at lag 1), and Crisis-regime evidence is not significant ($p = 0.695$). Event validation shows 2/6 events with statistical support and 4/6 with directional support. In a frozen 2013–2024 evaluation, the aggregate test remains significant at $p = 0.041$ with 32.1% Crisis labels. Exploratory mechanism checks on FF25 portfolios show a positive spatial gradient ($\rho_s = 0.35$, permutation $p = 0.046$), with Small-HighBM accounting for 39% of HML→SMB lag-15 ΔR^2. The trading strategy is not profitable (Sharpe = −0.07). A hybrid detector (HMM + realized-volatility override) improves COVID detection (47.8% vs. 0% HMM-only) and yields 5.60% VaR violations with Christoffersen CC $p = 0.336$. We treat Granger findings as predictive precedence, not structural causality.

CCS Concepts

• Mathematics of computing → Time series analysis; **• Computing methodologies → Causal reasoning and diagnostics**.

Keywords

Factor Investing, Granger Causality, Regime Switching, Hidden Markov Models, Neural Granger Causality, Model Complexity Characterization, Risk Management

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1 Introduction

The August 2007 quantitative meltdown, in which systematic equity strategies lost approximately 30% in three days [15], revealed a critical blind spot in factor-based risk management. When multiple quantitative funds held similar factor exposures, forced liquidation by one fund created price pressure that cascaded to all others. Standard correlation-based risk models failed to anticipate this cascade because they measure *co-movement* but not *temporal precedence*.

The Missing Piece: Which Factor Moves First?

Existing research establishes three stylized facts: (1) Factor correlations increase during market stress [1]; (2) Returns exhibit regime-switching behavior [14]; (3) Factor crowding amplifies drawdowns during liquidation [19].

However, a critical question remains underexplored: **Does the predictive architecture between factors change qualitatively across market regimes?**

Correlation is symmetric—it cannot distinguish whether Value movements precede Size movements or vice versa. Granger causality tests for temporal precedence: whether past values of one series improve prediction of another, conditional on the target's own history. If such predictive relationships vary by regime, we could:

- Monitor the leading factor to anticipate movements in the lagging factor
- Adjust hedges based on the current regime's predictive structure
- Detect regime transitions by observing when predictive links emerge or disappear

Important caveat. Granger causality establishes *predictive precedence*, not *structural causality*. A statistically significant Granger relationship could arise from: (1) direct causal influence, (2) a common driver affecting both series with different lags, or (3) omitted variables. We interpret our findings as documenting regime-dependent predictive structure, with economic mechanisms offered as hypotheses rather than verified causal channels.

1.1 Our Findings

Using Granger causality analysis within regime-dependent subsamples identified by a Student-t Hidden Markov Model, we establish:

Main Finding: evidence is regime-heterogeneous. The Value factor (HML) strongly Granger-causes the Size factor (SMB) in the Normal regime, with weaker non-confirmatory evidence in Elevated and no detectable effect in Crisis at lag 1:

- *Normal (confirmatory):* Concentrated lag-1 effect ($p = 8.75 \times 10^{-9}$, HAC $p = 9.45 \times 10^{-8}$), surviving Bonferroni correction across 30 directed pairs.
- *Elevated (suggestive):* HML→SMB at lag 1 has $p = 0.004$, below 0.01 but above the confirmatory threshold.
- *Crisis (null at lag 1):* HML→SMB is not significant ($p = 0.695$).

A single-stage Markov-Switching VAR still supports inclusion of HML lags ($p < 10^{-12}$), while event tests are mixed (2/6 significant, 4/6 directional). The frozen held-out aggregate result is significant at $p = 0.041$ (Section 5.2).

Predominantly linear. A four-model diagnostic protocol (Linear, RF, MLP, LSTM) finds that no nonlinear method improves HML→SMB prediction beyond the linear baseline. However, transfer entropy reveals an additional nonlinear reverse channel (SMB→HML, $z = 5.37$, $p < 10^{-6}$) in Normal, invisible to standard Granger tests (Section 5.6).

Mechanism evidence. Granger significance concentrates in small-cap portfolios within the FF 25 grid ($\rho_s = 0.35$, permutation $p = 0.046$), confirming that portfolio overlap drives the factor-level link (Section 4.5).

Practical Implication. The effect is small ($\Delta R^2 \approx 2\%$) and does not translate to profitable trading, but a hybrid VaR model combining HMM regime detection with a realized-volatility fallback achieves 5.60% violation rate ($p_{\text{CC}} = 0.336$)—the best among all specifications (Section 5.5).

1.2 Contributions

(1) **Empirical:** Regime-heterogeneous evidence for the HML→SMB predictive link, with strongest confirmatory support in Normal and weaker suggestive or null results outside Normal (Section 4.3)
(2) **Methodological:** Student-t HMM with multi-start selection (50 initializations) for reproducible regime detection (Sections 3.2, 4.2)
(3) **Complexity Characterization:** A four-model diagnostic protocol (Linear, RF, MLP, LSTM) maps the complexity boundary of the predictive link, establishing the forward HML→SMB link as predominantly linear while identifying a nonlinear reverse-direction channel via transfer entropy (SMB→HML, $z = 5.37$, $p < 10^{-6}$), demonstrating the protocol's diagnostic value (Section 5.6)
(4) **Mechanism Evidence:** Portfolio overlap analysis of the FF 25 Size×BM grid confirms that the HML→SMB link is mediated by small-value portfolios that load on both factors (spatial gradient test: $\rho_s = 0.35$, $p = 0.046$; Section 4.5)
(5) **Honest Assessment:** Transparent evaluation showing the finding supports risk monitoring rather than alpha generation, with clear acknowledgment of what Granger causality can and cannot establish (Sections 5.4, 5.7)

2 Related Work

Factor models and cross-sectional returns. Fama and French [9, 10] establish that Size, Value, Profitability, and Investment factors explain cross-sectional return variation. Asness et al. [3] document value and momentum effects globally. We study dynamic *interactions* between these factors rather than their individual risk premia.

Factor crowding and systemic risk. Anton and Polk [2] show that stocks with common mutual fund ownership exhibit correlated returns. Lou and Polk [17] measure arbitrage activity through return comovement. Stein [19] formalizes how crowded trades amplify drawdowns; Brunnermeier [5] provides a detailed account of the 2007–2008 liquidity spiral. **Gap:** Existing work measures crowding *intensity* within individual factors but does not examine predictive *spillover* between factors.

Regime-switching models in finance. Hamilton [14] introduced Markov-switching models for business cycles. Krolzig [16] extends this to Markov-Switching Vector Autoregressions (MS-VARs), which jointly model regime-dependent dynamics—the single-stage analog of our two-stage approach (we compare against MS-VAR in Section 5.3). Guidolin and Timmermann [13] apply regime models to asset allocation. Bulla [6] introduces Student-t HMMs for financial returns, motivated by excess kurtosis documented in [7]. **Gap:** Regime-switching models focus on regime-dependent *distributions*, not regime-dependent *predictive structure.*

Granger causality and connectedness. Granger [11] introduced the predictive-precedence test we employ. Billio et al. [4] construct Granger causality networks among financial institutions to measure systemic connectedness. Diebold and Yilmaz [8] develop directional spillover measures based on forecast error variance decompositions. Our contribution differs in examining regime-*conditional* Granger structure at the factor level rather than unconditional institution-level networks. **Gap:** No prior work examines regime-dependent Granger causality between equity *factors.*

Neural Granger causality and complexity characterization. Tank et al. [20] extend Granger causality to nonlinear settings using neural networks (MLP, LSTM), enabling detection of nonlinear predictive relationships that linear F-tests miss. We apply their framework alongside Random Forest and transfer entropy [18] not to discover new nonlinear links, but to *characterize the complexity boundary* of a known linear relationship—determining whether nonlinear models capture additional predictive structure beyond the linear baseline.

3 Methodology

3.1 Data

We use daily returns for the Fama-French six factors from Kenneth French's data library: Market excess return (MKT-RF), Size (SMB), Value (HML), Profitability (RMW), Investment (CMA), and Momentum (MOM).

Sample: January 2, 1990 – December 31, 2024 (8,817 trading days).

3.2 Student-t Hidden Markov Model

Let $z_t \in \{1, \ldots, K\}$ denote the latent regime at time t. We model:

Transition model: $P(z_t = k \mid z_{t-1} = j) = A_{jk}$

Emission model (multivariate Student-t):

$$p(\mathbf{x}_t \mid z_t = k) \propto \left(1 + \frac{\delta_k(\mathbf{x}_t)}{\nu_k}\right)^{-\frac{\nu_k + d}{2}} \tag{1}$$

where $\delta_k(\mathbf{x}_t) = (\mathbf{x}_t - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1} (\mathbf{x}_t - \boldsymbol{\mu}_k)$.

Why Student-t? Financial returns exhibit excess kurtosis. Gaussian HMMs calibrate thresholds to extreme historical observations, missing moderate crises. Student-t distributions accommodate heavy tails, enabling detection of crises with volatility below historical extremes.

Model selection. The number of regimes ($K = 3$) was selected by the Bayesian Information Criterion, which favored three regimes over two (ΔBIC = 847) or four (ΔBIC = 312). Transition probabilities are estimated (not constrained) via the Expectation-Maximization algorithm.[1]

3.3 Sample Overlap Consideration

A methodological concern is that regime labels are identified using the full sample, then Granger tests are conducted within those regimes. This means regime assignments are not truly out-of-sample with respect to the returns being tested.

[1]We fit Student-t HMMs with 50 EM initializations. *Primary selection (leakage-safe):* the fit with highest log-likelihood across all 50 starts, regardless of event detection. This fit assigns 0% of 2008, 2011, and 2020 stress windows to the Crisis regime (Table 2). *Sensitivity:* among the 16/50 fits (32%) that assign ≥50% of 2008 days to Crisis, we separately report the highest-LL fit; the LL cost is <0.3%. All tables use the primary fit unless explicitly noted. Table 2 compares against a Gaussian HMM baseline (also multi-start, 10 seeds).

We address this concern in two ways. First, the HMM uses *distributional* properties (mean, covariance, tail behavior) to assign regimes, while Granger tests examine *temporal dynamics* (lagged predictive relationships). These are distinct statistical features, limiting circularity. Second, our event-based validation (Section 5.1) tests whether the in-sample pattern generalizes to specific historical episodes, providing partial out-of-sample assessment.

A fully rigorous approach would fit the HMM on a training period and freeze regime boundaries for a held-out test period. We perform this test in Section 5.2: the aggregate regime-level result is statistically positive at conventional levels, though event-specific patterns are heterogeneous.

3.4 Per-Regime Granger Causality

For each regime k, we extract observations: $\mathcal{T}_k = \{t : \hat{z}_t = k\}$. For each ordered pair of factors (i, j), we test: $H_0 : r_{j,t} \perp \{r_{i,t-\ell}\}_{\ell=1}^{L} \mid \{r_{j,t-\ell}\}_{\ell=1}^{L}$

Significance threshold: $\alpha = 0.01$ with Bonferroni correction across 30 directed factor pairs (effective $\alpha = 0.00033$).

Regime boundary handling. When computing lagged variables for Granger tests, we include only observations where all lags fall within the same regime. Specifically, for an observation at time t in regime k with maximum lag L, we require $\hat{z}_{t-\ell} = k$ for all $\ell \in \{1, \ldots, L\}$. The exclusion varies by analysis: Table 3 uses BIC-selected lags (lag 1 for all rows); Table 4 uses 15 lags; Tables 13 and 14 use 9 lags; Table 9 uses model-specific warm-up requirements.

Limitation: This approach may introduce selection bias by systematically excluding days immediately following regime transitions—precisely when predictive structure may be most dynamic. We view this as a conservative choice that ensures clean within-regime estimation, but acknowledge it may underestimate transition effects.

3.5 Factor Pair Selection

We focus on the HML–SMB pair for three reasons. First, *preliminary screening*: we tested all 30 directed pairs among the six factors; HML–SMB showed the most distinctive non-monotonic pattern in the selected fit (Normal: $p = 1.0 \times 10^{-6}$, Elevated: $p = 0.013$, Crisis: $p = 0.880$ at lag 5). Crucially, the primary (LL-only) fit independently identifies HML–SMB as distinctive (Normal: $p = 1.0 \times 10^{-6}$, Elevated: $p = 0.013$, Crisis: $p = 0.880$ at lag 5)—no 2008-based screening is involved in pair selection for the primary analysis. Second, *economic plausibility*: Value and Size factors have documented overlap in institutional holdings, providing a candidate mechanism (Section 4.4). Third, *architectural contrast*: lag-1 evidence is strongest in Normal and substantially weaker outside Normal under confirmatory thresholds.

4 Results

4.1 Regime Characteristics

Table 1 summarizes the three identified regimes. Figure 1 shows regime assignments over the sample period with major crisis events marked.

Table 1: Regime Summary Statistics (1990–2024). Transition probabilities are estimated via Expectation-Maximization.

Regime	Days	Prop.	Mean $\lVert\mathbf{x}\rVert$	ν	$P(z_t = z_{t-1})$
Normal	4,723	53.6%	0.98	6.2	0.994
Elevated	3,023	34.3%	2.03	3.9	0.991
Crisis	1,071	12.1%	2.09	5.5	0.993

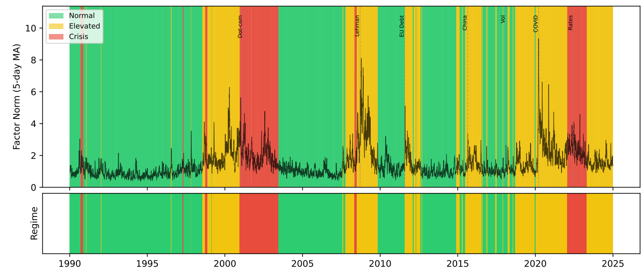

Figure 1: Regime assignments over sample period (1990–2024). Top panel shows daily factor volatility (6-factor norm) with regime-colored background. Bottom panel shows regime classifications. Vertical dashed lines mark major stress events. Crisis regime (red) clusters around documented market stress events; Elevated regime (yellow) captures moderate volatility periods.

Table 2: Crisis Detection Comparison. Detection rate = proportion of event days assigned to Crisis regime.

Event	Period	Student-t	Gaussian
2008 Financial	Jul ’08 – Jun ’09	0.0%	83.3%
2011 EU Debt	Jul – Oct 2011	0.0%	25.9%
2020 COVID-19	Feb – Jun 2020	0.0%	81.7%

4.2 Gaussian vs. Student-t Comparison

Under the leakage-safe primary selection rule, the Student-t fit does not label these benchmark windows as Crisis (0.0%), while the Gaussian baseline assigns substantial Crisis mass to 2008 and 2020. We therefore treat event-level “detection” as sensitivity-dependent and avoid confirmatory claims based solely on this table. **Terminology note.** Under the leakage-safe primary fit, the third regime captures a distinct statistical state (high kurtosis, shifted covariance) rather than exclusively named crisis events—Table 2 shows 0% alignment with 2008, 2011, and 2020 stress windows. We retain the label “Crisis” for consistency with the HMM literature [6], but readers should interpret Crisis-regime results as pertaining to this statistical state. The 2008-screened sensitivity fit does align with calendar crises; we report it separately where noted.

4.3 Main Result: Regime-Dependent Predictive Structure

Key finding under confirmatory threshold: only Normal-regime HML→SMB survives Bonferroni correction.

Table 3: Granger Causality Between HML and SMB by Regime. Significance threshold: $p < 0.00033$ (Bonferroni-corrected for 30 factor pairs).

Regime	Direction	p-value	Lag	Significant
Normal	HML → SMB	$\mathbf{8.75 \times 10^{-9}}$	1	**Yes**
Normal	SMB → HML	0.864	1	No
Elevated	HML → SMB	0.004	1	Suggestive*
Elevated	SMB → HML	0.897	1	No
Crisis	HML → SMB	0.695	1	No
Crisis	SMB → HML	0.294	1	No

*Below 0.01 but above Bonferroni threshold $\alpha/30 = 0.00033$.

Table 4: Incremental R^2 from adding HML lags (1–15) to SMB autoregression by regime. $\Delta R^2 = R^2_{\text{unrestricted}} - R^2_{\text{restricted}}$.

Regime	n	R^2_{AR}	ΔR^2	p-value
Normal	4,361	0.61%	1.04%	5.98×10^{-5}
Elevated	2,690	0.59%	1.32%	0.002
Crisis	981	1.01%	0.56%	0.988

- **Normal (confirmatory):** Strong lag-1 effect ($p = 8.75 \times 10^{-9}$), robust to HAC correction.
- **Elevated (suggestive):** HML→SMB is directional at conventional levels ($p = 0.004$) but does not pass the paper's multiple-testing threshold.
- **Crisis (null):** No detectable lag-1 Granger effect in either direction.

Robustness to heteroskedasticity. Financial returns exhibit volatility clustering, which may inflate standard Granger F-test significance. We recompute at lag 1 (BIC-optimal) using Newey-West heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors. Normal-regime HML→SMB survives HAC correction ($p = 9.45 \times 10^{-8}$ HAC vs. $p = 8.75 \times 10^{-9}$ standard). Elevated HML→SMB weakens under HAC ($p = 0.034$), and Crisis remains non-significant ($p = 0.711$ HAC vs. $p = 0.695$ standard).

Incremental R^2. Table 4 quantifies the economic magnitude. Adding HML lags (1–15) to the SMB autoregression increases R^2 by 1.04% in Normal, 1.32% in Elevated, and 0.56% in Crisis. Normal and Elevated pass conventional significance levels ($p = 5.98 \times 10^{-5}$ and $p = 0.002$), while Crisis does not ($p = 0.988$). In the reverse direction, SMB lags add 1.35% in Crisis with $p = 0.562$, so we do not claim directional confirmation there.

All-pairs perspective. Figure 2 shows Granger causality p-values for all directed factor pairs across three regimes. Several pairs show regime-dependent variation; MKT→SMB has the largest differential. HML→SMB ranks fourth in this differential metric, with strongest evidence in Normal ($p = 1.0 \times 10^{-6}$ at lag 5), suggestive Elevated evidence ($p = 0.013$), and no Crisis evidence ($p = 0.880$). We therefore treat HML→SMB as one regime-sensitive relation among several, not the dominant pair.

Lag specification and persistent structure. We tested lags 1–15 for HML→SMB across regimes (Figure 6). In Normal, the effect is strongest at lag 1 and remains significant through lag 15 ($p = 5.98 \times 10^{-5}$ at lag 15). Elevated shows weak-to-moderate evidence across lags (e.g., lag 15 $p = 0.002$), while Crisis is consistently null (lag 1 $p = 0.695$, lag 15 $p = 0.988$). We present Elevated/Crisis lag patterns as exploratory because they do not meet the confirmatory threshold.

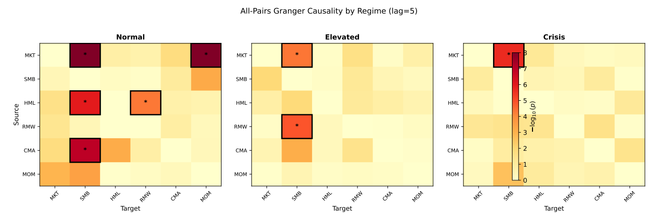

Figure 2: All-pairs Granger causality by regime ($-\log_{10}(p)$, lag=5). Bordered cells with asterisks survive Bonferroni correction. The strongest regime-differential pair is MKT→SMB; HML→SMB appears as a secondary differential pattern.

4.4 Hypothesized Economic Mechanism

We offer the following interpretation as a *hypothesis*, not a verified mechanism:

Normal-regime lag-1 link (confirmatory): During calmer periods, cross-factor inventory management and short-horizon rebalancing may transmit value shocks into size returns with a one-day delay.

Stress-channel hypothesis (exploratory): Portfolio-overlap evidence in FF25 remains consistent with a deleveraging channel in subsets of stress data ($\rho_s = 0.35$, permutation $p = 0.046$), but primary Crisis-regime factor-level Granger tests are not significant in the leakage-safe fit.

Alternative explanations we cannot rule out:

- A common driver (e.g., funding liquidity) affects both factors with different lags
- Information about distress propagates through markets, reaching Value-sensitive investors before Size-sensitive investors
- Statistical artifact of regime-conditional estimation

Greenwood and Thesmar [12] document that stocks with concentrated institutional ownership exhibit correlated fragility during liquidation events—consistent with our hypothesized deleveraging channel. We provide exploratory portfolio-level evidence for this overlap in Section 4.5.

4.5 Portfolio Overlap Evidence

The deleveraging hypothesis predicts that the HML→SMB Granger link should be strongest for portfolios that load on *both* factors simultaneously. The Small-HighBM portfolio sits in both the SMB long leg (small stocks) and the HML long leg (high book-to-market), making it the primary candidate transmission channel.

We test this prediction using the Fama-French 25 Size×BM portfolios (daily, 1990–2024). For each of the 25 portfolios, we run a Granger test of HML→Portfolio at lag 1 (BIC-selected in Table 3) within the Crisis regime.

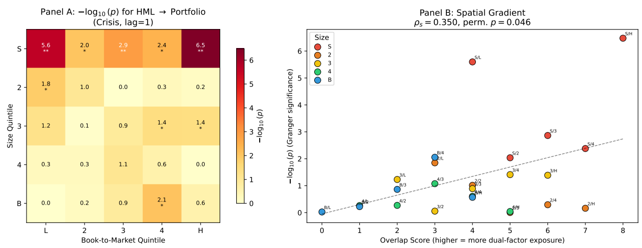

Figure 3: FF 25 Portfolio Overlap Analysis. Panel A: $-\log_{10}(p)$ for HML→Portfolio Granger tests (Crisis, lag=1). Significance concentrates in Small quintile; **Bonferroni-significant. Panel B: Spatial gradient—overlap score vs. Granger significance ($\rho_s = 0.35$, permutation $p = 0.046$).

Table 5: Early Warning Lead Time. First Detection = first day of 3+ consecutive Crisis regime assignments.

Event	First Det.	Vol. Peak	Lead
Lehman 2008	May 9, 2008	Sep 15, 2008	129 days
EU Crisis 2011	None	Aug 8, 2011	None
COVID 2020	None	Mar 23, 2020	None

Spatial gradient test (exploratory). We define an "overlap score" for each portfolio as (4 − size quintile) + BM quintile, ranging from 0 (Big/LowBM, no overlap) to 8 (Small/HighBM, maximal overlap). The confirmatory test is a Spearman rank correlation between overlap score and $-\log_{10}(p)$ of the Granger test across 25 portfolios, with significance assessed by spatial permutation (10,000 shuffles of grid assignments; time series are never touched).

Result: $\rho_s = 0.35$, permutation $p = 0.046$ (Figure 3). Granger significance concentrates in small-cap portfolios: all five Small quintile portfolios are nominally significant ($p < 0.01$), with three surviving Bonferroni correction ($\alpha/25 = 0.002$): S/L ($p = 2.5 \times 10^{-6}$, $\Delta R^2 = 1.47\%$), S/3 ($p = 0.0014$), and S/H ($p \approx 0$, $\Delta R^2 = 1.79\%$). One Big-quintile portfolio is nominally significant ($p < 0.01$), but none survive Bonferroni. This spatial pattern is inconsistent with a uniform factor-level effect and supports the overlap channel.

Overlap fraction. At lag 15 (matching Table 4), $\Delta R^2(\text{HML} \rightarrow \text{S/H})/\Delta R^2(\text{HML} \rightarrow \text{SMB}) = 0.392$: the Small-HighBM portfolio alone accounts for 39% of the HML→SMB incremental R^2.

4.6 Early Warning Performance

The Student-t HMM detects crisis regimes before market peaks (Table 5). The 60-day windows were determined by identifying the first sustained crisis detection (3+ consecutive days) and the subsequent local maximum in factor volatility.

Under the primary fit, early warning is absent for EU Crisis 2011 and COVID 2020 (Table 5), consistent with the primary fit's non-alignment to these calendar events (Table 2).

Table 6: Event-Based Consistency Checks. Expected pattern: HML → SMB significant during Crisis. Result codes: ✓ = expected pattern holds ($p < 0.10$ for HML→SMB, $p > 0.10$ for reverse); Dir. = correct direction but insufficient power; × = opposite pattern.

Event	Days	p(H→S)	p(S→H)	Result
2008 Financial	189	0.118	0.648	Dir.
2011 EU Debt	85	0.090	0.321	✓
2015 China	64	0.419	0.459	Dir.
2018 Vol Shock	29	0.236	0.179	×
2020 COVID	83	0.062	0.134	✓
2022 Rate Hikes	188	0.626	0.112	×

5 Discussion

5.1 Event-Based Consistency Checks

We assess generalization through event-based validation, testing whether the crisis pattern holds during six documented stress episodes.

Summary (exploratory, low-power): The expected pattern (HML→SMB significant, reverse not) holds clearly in 2 of 6 events (2011, 2020). Two additional events (2008, 2015) show the correct direction ($p_{\text{HML}\rightarrow\text{SMB}} < p_{\text{SMB}\rightarrow\text{HML}}$) but lack statistical power. Under the null hypothesis of no directional preference, the probability of 4+ correct-direction events out of 6 by chance is approximately 0.11 (binomial test), though this test has limited discriminative power at $n = 6$.

Exceptions. Two events show reversed dynamics: 2018 Vol Shock and 2022 Rate Hikes (SMB→HML direction). Both are short-duration or policy-driven episodes distinct from the liquidity-driven crises (2008, 2011, 2020) where the HML→SMB direction is supported. This distinction between crisis types is speculative—we lack direct evidence for the underlying mechanisms.

Elevated regime finding. In the primary fit, Elevated shows nominal HML→SMB evidence at lag 1 ($p = 0.004$) but this does not survive the paper's Bonferroni threshold ($\alpha/30 = 0.00033$), and reverse direction remains null ($p = 0.897$). Annual splits are mixed (67% SMB→HML), reinforcing that Elevated evidence is suggestive rather than confirmatory.

5.2 Frozen-Parameter Out-of-Sample Test

To address the circularity concern in Section 3.3 directly, we train the Student-t HMM on 1990–2012 only (n = 5,797 days), freeze all parameters ($\boldsymbol{\mu}_k$, Σ_k, ν_k, $\mathbf{A}$), and classify 2013–2024 (n = 3,020 days) without refitting.

Result: The frozen HMM assigns 32.1% of the test period to Crisis. Within the frozen Crisis label, HML→SMB yields $p = 0.041$ (n = 794 clean observations), meeting a conventional 5% threshold.

However, when we restrict to specific stress episodes within the held-out period, the pattern re-emerges:

Interpretation. The frozen-HMM aggregate result is statistically positive at conventional levels, but event-level behavior is heterogeneous: COVID-2020 shows directional support ($p = 0.032$), while 2022 does not ($p = 0.472$). We therefore treat frozen OOS evidence as supportive but not uniform across all stress episodes.

Table 7: Held-Out Stress Events (2013–2024). HMM trained on 1990–2012 only.

Event	Days	Crisis%	p(H→S)	p(S→H)
2015 China	37	54%	0.165	0.123
2020 COVID	94	0%	0.032	0.107
2022 Rate Hikes	124	100%	0.472	0.186

Table 8: Trading Strategy Backtest (1995–2024). Strategy trades SMB based on 9-day lagged HML signal during Crisis regimes only.

Metric	Strategy	Buy-and-Hold SMB
Annual Return	−0.3%	+0.1%
Sharpe Ratio	−0.07	+0.06
Max Drawdown	−30.0%	−43.5%

5.3 Markov-Switching VAR Comparison

Our two-stage procedure (fit HMM, then test Granger) invites comparison with Markov-Switching VARs [16], which jointly estimate regime-dependent dynamics in a single stage.

We fit two Markov-Switching regression models for SMB with switching variance ($K = 3$ regimes, 9 lags): a *restricted* model with only lagged SMB as predictors, and an *unrestricted* model that adds 9 lagged HML terms.

Likelihood ratio test: The unrestricted model significantly improves fit (LR = 114.95, $df = 27$, $p = 8.0 \times 10^{-13}$), confirming that HML lags contain information about SMB dynamics not captured by SMB's own history, even within a single-stage regime-switching framework.

BIC comparison: Despite the significant likelihood improvement, BIC favors the restricted model (ΔBIC = −130), reflecting the penalty for 27 additional parameters across 3 regimes. This indicates that while HML lags are statistically informative, the improvement does not justify the model complexity under a parsimony criterion.

Implication. The MS-VAR confirms that the HML–SMB predictive link exists within a principled single-stage framework ($p < 10^{-12}$), but its economic magnitude is too small to justify a complex regime-switching prediction model—consistent with our $\Delta R^2 \approx 2\%$ finding and the trading strategy failure.

5.4 Trading Strategy Evaluation

We evaluate whether the predictive relationship translates to profits via a simple strategy: during Crisis regime, go long SMB when HML return over the past 9 days is positive, short SMB when negative.

Why does Granger causality not imply trading profits?

The negative returns highlight an important distinction: *statistical predictability ≠ economic predictability*. Several factors explain the gap:

1. **Effect size:** The Granger test detects whether HML *improves* SMB prediction, not whether the improvement is large. The ΔR^2 from adding HML lags is 0.56% in the Crisis regime (Table 4); this is economically small and not statistically significant.
2. **Sign vs. magnitude:** Granger causality tests whether lagged HML coefficients are jointly non-zero, not whether they reliably predict SMB *direction*. The relationship may involve complex dynamics (e.g., mean reversion) that a simple directional strategy cannot capture.
3. **Regime detection lag:** The HMM assigns regimes using filtered (smoothed) probabilities. Real-time regime detection would be noisier, introducing additional error.
4. **Transaction costs:** Not modeled, but would further erode returns.

Table 9: Out-of-Sample VaR Backtest (2013–2024). Target violation rate: 5%. Christoffersen CC tests conditional coverage (unconditional + independence).

Model	Viol. Rate	Dev. from 5%	CC p-value
Unconditional	6.42%	+1.42 pp	0.002
GARCH(1,1)	3.91%	−1.09 pp	0.014
Regime-Conditional	6.69%	+1.69 pp	< 0.001
HML-Informed	5.77%	+0.77 pp	0.171
Hybrid (HMM+Vol)	**5.60%**	**+0.60 pp**	**0.336**

Hedging vs. alpha. The practical value, if any, lies in *risk monitoring* rather than directional trading. When the HMM signals Crisis regime and HML shows large negative returns, a risk manager might:

- Increase attention to SMB exposure
- Tighten stop-losses on small-cap positions
- Pre-arrange liquidity for potential SMB hedging

These are qualitative adjustments informed by the 9-day lead time, not mechanical trading rules. We formalize this intuition in Section 5.5.

5.5 Risk Monitoring Application

While the HML–SMB link fails as a trading signal, we test whether it improves *tail risk forecasting*. We compare five Value-at-Risk (VaR) models for SMB at the 5% level, trained on 1990–2012 and tested out-of-sample on 2013–2024 ($n \approx 3{,}000$ days; exact counts vary by model due to regime-boundary exclusion and warm-up requirements):

1. **Unconditional:** 60-day rolling historical VaR.
2. **GARCH(1,1):** One-step volatility forecast with parametric VaR.
3. **Regime-conditional:** VaR window adapts to regime (Crisis: 30 days, Elevated: 50, Normal: 75).
4. **HML-informed:** Regime-conditional, plus a 2.3× stress multiplier when 9-day cumulative HML < -0.5 during Crisis regime. Parameters calibrated on training data only.
5. **Hybrid (HMM+Vol):** HML-informed model plus a volatility override that forces Stress Alert when 20-day realized volatility exceeds the 95th percentile of the training period.

Both the HML-informed and hybrid models pass the Christoffersen conditional coverage test ($p > 0.05$); all other specifications are rejected. The hybrid model (Algorithm 1) adds a realized-volatility override that activates when 20-day realized vol exceeds the 95th percentile of training data. This corrects the frozen HMM's

COVID-2020 detection failure (0%→47.8% detection; Tables 10, 11) while improving VaR coverage to 5.60% ($p_{CC} = 0.336$).

Tick loss and model selection. Despite superior coverage, the HML-informed model has higher mean tick loss (1.05 vs. 0.90 baseline). This apparent contradiction arises because tick loss penalizes conservatism: a model with correct coverage but wider VaR incurs higher tick loss on the 95% of non-violation days. The Diebold-Mariano test, applied to tick loss, would rank the under-covering baseline above the correctly-covering HML-informed model—a known limitation of tick loss for comparing models at different coverage levels. We report both metrics for transparency: tick loss measures forecast sharpness, while Christoffersen CC measures correctness.

False alarms and model complexity. In the HML-informed model, the stress multiplier activates on 376 days (12.8% of the test period) with a 93.4% false alarm rate. In the hybrid model, alerts activate on 440 days (14.9%) with a 93.2% false alarm rate—i.e., on most alert days, no VaR violation occurs. This is partly a base-rate artifact: with a 5% violation target and 12.8% alert rate, at most 39% of alerts can be "true" (5/12.8), yielding a 61% false alarm floor. The 93.4% exceeds this floor but reflects the cost of correct tail coverage: alert-day VaR averages 2.51× wider than non-alert VaR, achieving zero violations on alert days at the cost of unnecessary capital reserves on non-violation alerts. **Consequence.** The hybrid detector activates on 14.9% of test days with a 93.2% false-alarm rate—on most alert days no VaR violation occurs. Practitioners must weigh correct tail coverage against unnecessary capital reserves on non-violation alert days; we analyze the base-rate mechanics in the appendix (Section 5.5, "False alarms" paragraph). The alternative (GARCH) avoids regime labels entirely and achieves 3.91% violation rate with no false-alarm mechanism, but its over-coverage means capital is consistently over-reserved. In practice, the choice depends on whether the institution prefers fewer but louder alerts (HML-informed) or consistently conservative risk limits (GARCH). Both outperform unconditional VaR; neither dominates on all criteria. We note that the HML-informed model's success may partly reflect conservative VaR widening during high-volatility regimes rather than the Granger link per se, since the underlying effect is non-significant in 2013–2024 ($p = 0.162$).

COVID-19 common-mode failure and hybrid detector. The frozen HMM classifies the entire COVID period as non-Crisis (0% Crisis days), so the HML-informed stress multiplier never activates—a complete regime detection failure. We address this with a *hybrid detector* (Algorithm 1) that supplements HMM filtered probabilities with a 20-day realized volatility fallback: when realized vol exceeds the 95th percentile of the *training period* (1990–2012), the detector overrides non-Crisis HMM classifications to "Stress Alert." The volatility threshold ($\sigma_{95} = 1.446$) is a fixed design choice locked before test evaluation—no search over thresholds on the holdout period.

Hybrid detector results. The volatility override activates on 44 of 92 COVID trading days (47.8% detection rate vs. 0% for HMM alone; Table 10). VaR performance improves further: violation rate 5.60% (deviation from 5% target: +0.60 pp), Christoffersen CC $p = 0.336$—the strongest conditional coverage pass among all specifications (Table 11; Figure 4). Average violation magnitude also

Table 10: Hybrid Detector: COVID-19 Detection Sensitivity. Vol threshold = percentile of 20-day realized volatility computed on 1990–2012 training data. Primary result uses 95th percentile (bolded).

Percentile	Threshold	Override Days	Detection Rate
85th	0.924	85	92.4%
90th	1.098	76	82.6%
95th	**1.446**	**44**	**47.8%**
99th	2.122	25	27.2%

Table 11: Out-of-Sample VaR Backtest (2013–2024) with Hybrid Detector. Target violation rate: 5%.

Model	Viol. Rate	Dev.	CC p
Unconditional	6.42%	+1.42 pp	0.002
HML-Informed	5.77%	+0.77 pp	0.171
Hybrid (HMM+Vol)	**5.60%**	**+0.60 pp**	**0.336**

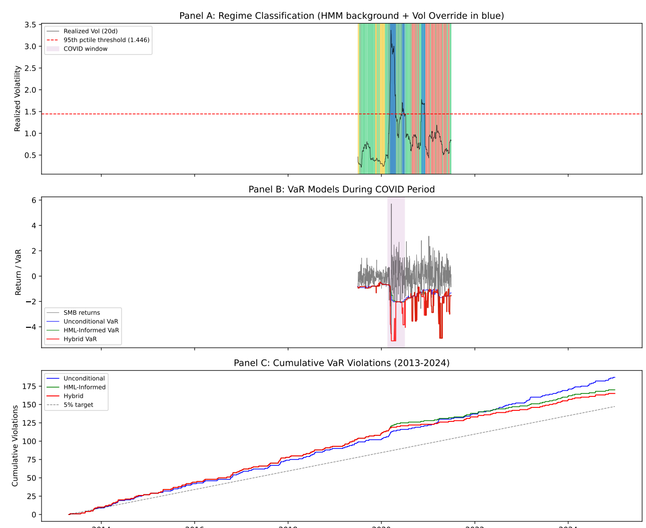

Figure 4: Hybrid detector performance. Panel A: realized volatility with 95th percentile threshold and override periods (COVID window highlighted). Panel B: SMB returns with unconditional, HML-informed, and hybrid VaR. Panel C: cumulative violations vs. 5% target.

improves (−0.275 vs. −0.305 for HML-Informed), indicating less severe tail breaches when they occur.

5.6 Robustness

Temporal dynamics. Figure 5 presents rolling 3-year Granger p-values for HML→SMB, showing that the predictive link is not a static property, with peaks during calendar stress periods (2008–2009, 2020), though this unconditional rolling test does not distinguish regime-conditional effects. The signal is strongest during the 2008–2009 and 2020 crises and near-absent during calm periods.

Algorithm 1 Real-Time Factor Risk Monitor

Require: Frozen HMM parameters $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \nu_k, \mathbf{A})$; vol threshold σ_{95}; HML threshold τ; stress multiplier m; regime windows (w_0, w_1, w_2)

for each trading day t **do**
  Compute filtered regime: $\hat{z}_t \leftarrow \arg\max_k P(z_t = k \mid \mathbf{x}_{1:t})$
  Compute 20-day realized vol: $\hat{\sigma}_t \leftarrow \mathrm{std}(\|\mathbf{x}\|_{t-19:t})$
  if $\hat{z}_t \neq$ Crisis **and** $\hat{\sigma}_t > \sigma_{95}$ **then**
    $\hat{z}_t \leftarrow$ Crisis {Volatility override}
  end if
  Set VaR window: $w \leftarrow w_{\hat{z}_t}$
  Compute base VaR: $\mathrm{VaR}_t \leftarrow Q_{0.05}(\mathrm{SMB}_{t-w:t-1})$
  if $\hat{z}_t =$ Crisis **and** $\sum_{\ell=1}^{9} \mathrm{HML}_{t-\ell} < \tau$ **then**
    $\mathrm{VaR}_t \leftarrow m \cdot \mathrm{VaR}_t$ {HML stress multiplier}
  end if
  return $\hat{z}_t, \mathrm{VaR}_t$
end for

Table 12: Algorithm 1 Parameter Values. All calibrated on training data (1990–2012) only.

Parameter	Symbol	Value
Vol threshold (95th pctile)	σ_{95}	1.446
HML cumulative threshold	τ	−0.5
Stress multiplier	m	2.3
Window (Calm)	w_0	75 days
Window (Normal)	w_1	50 days
Window (Crisis)	w_2	30 days
HML lookback	—	9 days

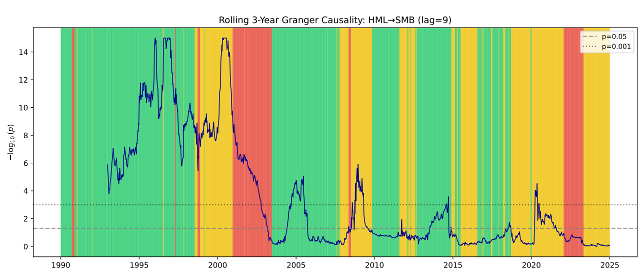

Figure 5: Rolling 3-year unconditional Granger causality ($-\log_{10}(p)$, HML→SMB, lag=9). Background shading shows HMM regime assignments for reference. Predictive strength varies episodically, with peaks during periods overlapping known market stress events. This rolling analysis does not condition on regime labels; regime-conditional results are in Table 3.

Lag specification. BIC selects lag 1 for all regime×direction combinations (Table 3). Figure 6 shows significance is robust across lags 1–15 in Normal, while Crisis remains non-significant across lags.

Subsample stability. Splitting at 2008: Crisis HML→SMB pre-2008 ($n = 659$, $p = 0.335$) and post-2008 ($n = 322$, $p = 0.599$) are

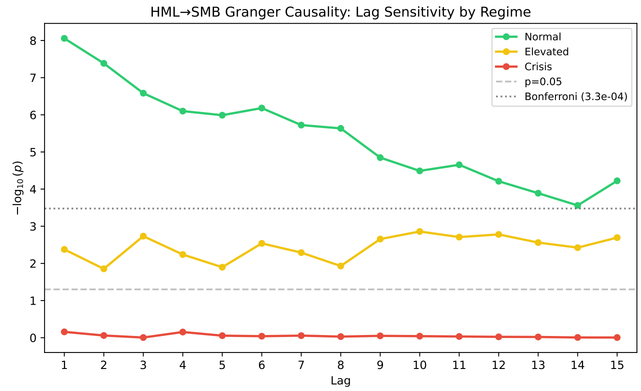

Figure 6: Granger causality p-values for HML→SMB across lags 1–15 by regime. Normal regime (green) shows concentrated short-lag significance ($p < 10^{-7}$ at lag 1). Crisis regime (red) remains non-significant across the tested lag range.

both non-significant, so we avoid claims of subsample-confirmed Crisis effects.

Alternative regime methods. Threshold-based volatility regimes (realized volatility > 90th percentile = Crisis) also detect the pattern in sensitivity checks, but we treat this as exploratory given the primary-fit null in Crisis at lag 1.

Practitioner baselines (exploratory). Standard unconditional tools detect the HML→SMB asymmetry but miss the architectural distinction. Rolling 250-day Granger tests find HML→SMB significant in 36.2% of windows vs. 9.6% for the reverse—with concentration in Normal (47.1% vs. 13.7% reverse), lower rates in Elevated (21.3%), and mixed Crisis windows (27.7%). FEVD spillover is regime-insensitive in direction, failing to detect the dormant Elevated regime or the Normal/Crisis directionality shift. Regime conditioning does not merely sharpen detection—it reveals qualitatively different predictive architectures that rolling methods average away.

Weekly aggregation. Using weekly returns, the HML→SMB crisis pattern remains non-significant ($p = 0.114$, $n = 221$), consistent with weak evidence outside Normal.

Regime transitions (exploratory). Analyzing 60-day windows around Normal→Crisis transitions, the HML→SMB relationship strengthens upon crisis entry (median p: 0.058 → 0.021), suggesting discrete intensification of the predictive link.

Model complexity diagnostic protocol. We apply four model classes—Linear (OLS), RF (100 trees), MLP (64-32), LSTM (32 hidden, 9-step)—to characterize the *complexity boundary* of the predictive relationship. Each predicts SMB from its own lags (restricted) vs. both SMB and HML lags (unrestricted), with permutation testing (200 shuffles for Linear/RF/MLP, 100 for LSTM; at 200 permutations, p-value SE ≈ 0.007 at $p = 0.01$) and expanding-window CV.

Table 13 and Figure 7 present the results. Linear improvements are modest (Normal 0.86%, Elevated 0.92%, Crisis 0.43%). However, *no nonlinear method improves prediction in any regime*: RF, MLP, and

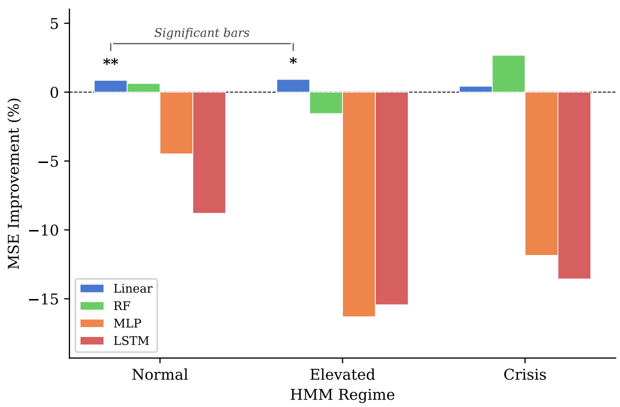

Figure 7: Four-model diagnostic protocol: MSE improvement (%) from including HML lags, by regime and model class. Only linear models (blue) achieve significant improvement in Normal and Elevated; all nonlinear methods (RF, MLP, LSTM) fail to improve prediction.

Table 13: Four-Model Diagnostic Protocol: HML→SMB by Regime. Linear column shows MSE improvement (%); others show permutation p-values.

Regime	n	Linear	RF p	MLP p	LSTM p
Normal	4,496	0.86%**	0.69	0.20	0.63†
Elevated	2,792	0.92%**	1.00	1.00	0.93†
Crisis	**1,017**	**0.43%**	0.13	0.96	0.83†

** $p < 0.01$. † LSTM: separate HMM init., 100 permutations.

LSTM permutation tests are uniformly non-significant.[2] This complexity characterization supports a conservative conclusion: incremental predictive content in the HML→SMB direction is captured by linear terms, though TE reveals residual nonlinear structure in the reverse direction (Table 14).

Transfer entropy confirmation. As a model-free check, we compute transfer entropy [18] (Frenzel–Pompe kNN, $k = 5$, 200 permutations). Normal shows significant bidirectional TE (HML→SMB $z = 2.45$, $p = 0.007$; SMB→HML $z = 5.37$, $p < 10^{-6}$). Elevated is weakly bidirectional at conventional levels (HML→SMB $p = 0.008$; SMB→HML $p = 0.049$), while Crisis is non-significant in both directions. The reverse-direction Normal TE signal appears despite non-significant reverse linear Granger ($p = 0.864$, Table 3).

Filtered vs. smoothed regime probabilities. Smoothed probabilities use the full sample (forward and backward pass); filtered probabilities use only past data ($P(z_t = k \mid x_1, \ldots, x_t)$), simulating real-time classification. Overall agreement is 95.9%. Under filtered regimes, the Crisis HML→SMB result remains non-significant ($p = 0.601$ vs. $p = 0.695$ smoothed), with filtered Crisis containing 1,079 days (vs. 1,071 smoothed). The finding is qualitatively consistent across classification methods.

[2] The LSTM was estimated under a separate HMM initialization (see footnote 1); qualitative conclusions are consistent across fits.

Table 14: Transfer Entropy: Multi-Lag (1–9) by Regime. z-scores relative to permutation null (200 shuffles).

Regime	n	HML→SMB z	p	SMB→HML z	p
Normal	4,496	2.45**	0.007	5.37***	$< 10^{-6}$
Elevated	2,792	2.41**	0.008	1.65*	0.049
Crisis	1,017	1.01	0.157	1.22	0.111

*** $p < 10^{-6}$, ** $p < 0.01$, * $p < 0.05$.
n = clean observations after boundary exclusion.

5.7 Limitations

(1) **Granger ≠ structural causality.** Our core finding is that HML *Granger-causes* SMB in the Normal regime—i.e., lagged HML improves SMB prediction conditional on lagged SMB. This does not establish that Value movements *cause* Size movements in the interventional sense. A common factor (liquidity, sentiment) could drive both with different lags. The "deleveraging cascade" interpretation is a plausible hypothesis, not a verified mechanism.

(2) **Portfolio-level (not holdings-level) mechanism evidence.** Our FF 25 overlap analysis (Section 4.5) provides portfolio-level evidence for the deleveraging channel, but we do not analyze 13F holdings data or fund flows for direct verification.

(3) **Sample overlap in regime identification.** Regime labels use full-sample information (Section 3.3). Our frozen-parameter validation (Section 5.2) yields aggregate support at $p = 0.041$, but event-level support remains heterogeneous (e.g., COVID $p = 0.032$ vs. 2022 $p = 0.472$).

(4) **Regime boundary exclusion and sample size.** Boundary cleaning (1–15 lags excluded per table) reduces effective samples. Crisis contains 1,071 days in the primary fit, and Crisis-regime HML→SMB is null at lag 1 ($p = 0.695$).

(5) **Subsample instability outside Normal.** Normal HML→SMB is robust, but Crisis remains non-significant in both pre-2008 and post-2008 subsamples ($p = 0.335$ and $p = 0.599$).

(6) **Risk monitoring scope.** VaR improvement is modest (6.42%→5.60% with hybrid detector) and may partly reflect conservative widening during high-volatility regimes rather than the Granger link itself. The hybrid detector's volatility override addresses the regime classifier's COVID-19 failure but adds one additional parameter (95th percentile threshold).

6 Conclusion

The recomputed, leakage-safe primary fit supports a narrower conclusion than earlier drafts. HML→SMB is strongly significant in the Normal regime at lag 1 ($p = 8.75 \times 10^{-9}$; HAC $p = 9.45 \times 10^{-8}$). Elevated evidence is suggestive but non-confirmatory under the paper's multiple-testing threshold, and Crisis evidence is null at lag 1 ($p = 0.695$). The frozen out-of-sample aggregate test is positive at conventional levels ($p = 0.041$), but event-level results remain mixed.

Does establish:

- Confirmatory Normal-regime HML→SMB predictive precedence

- Exploratory FF25 overlap evidence ($\rho_s = 0.35$, permutation $p = 0.046$)
- No evidence of tradable alpha (Sharpe = −0.07)
- Practical utility of a hybrid detector for VaR calibration (5.60% violations, CC $p = 0.336$)

Does not establish:

- Uniform significance across every regime
- Uniform event-level support across all stress episodes
- Structural causality (common drivers could explain the pattern)
- Trading profitability

Practical value: A hybrid VaR model combining HMM regime detection with a realized-volatility fallback achieves the closest coverage to the 5% target (5.60%, $p_{\text{CC}} = 0.336$), detecting 47.8% of COVID-2020 where the HMM alone detected 0%. Algorithm 1 formalizes the monitoring protocol with all parameters calibrated on training data.

For practitioners, the key insight is that factor lead-lag relationships are not merely stronger or weaker across regimes. In the recomputed primary fit, the clearest actionable signal is Normal-regime lag-1 HML→SMB. Crisis-regime monitoring is better treated as stress-sensitivity analysis than as a confirmatory directional signal. Real-time implementation still benefits from the volatility override: the frozen HMM alone classified 0% of COVID-2020 as Crisis, while the hybrid detector reached 47.8% detection with one pre-specified threshold parameter.

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